
1. 8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Each of (i), (ii), (vi) being well-defined collection is a set. Of course, (ii) is empty set, having no element. Also, (vi) has FOUR elements, and no repeated element. On the other hand, each of (iii) and (iv) is not well-defined, hence is NOT a set (Each of 'most renowned' or 'most talented' is not well-defined). (v) *is generally not considered as a set*, as repetition of elements in set notation is not allowed
- 2) a) $\{2, 3, 5\}$, cardinality = 3,
b) $\{5, -2\}$, cardinality = 2
- 3) a) it is a set; $\{x: x \in N, 3 \text{ divides } x, \text{ and } 6 \leq x \leq 27\}$, cardinality = 8
b) it is a set, $\{x: x = 2^n, n \in N\}$, cardinality = infinite (or, ∞)
c) it is a set, $\{x: x = \{1, \dots, i\}, i \in N, \text{ and } i \leq 10\}$, cardinality = 10.

Check Your Progress 2

- 1) i) $X=Y$,
ii) $X \neq Y$, as $15 \in Y$, and $15 \notin X$
iii) $X \neq Y$, as $2 \in X$, but $2 \notin Y$
iv) $X=Y$.
- 2) i) $X \subset Y$, as IGNOU has students enrolled in other disciplines also,
ii) $X \not\subset Y$ and $Y \not\subset X$,
iv) $X \subset Y$,
v) $X \not\subset Y$ as $2 \in X$ and $2 \notin Y$; $Y \not\subset X$ as $\{2, 3\} \in Y$ and $\{2, 3\} \notin X$
- 3) i) False
ii) False
iii) False
iv) True
v) True
vi) False
vii) True.

Check Your Progress 3

- 1) i) it is convex
ii) it is NOT convex
- 2) 8
- 3) i) There are $2^6 = 64$
ii) The answer is 62: there are $2^6 = 64$ subsets in U altogether. We exclude two: U itself (because it is not proper) and the empty set (because it is not non-empty).
- 4) $\{d\}, \{a, d\}, \{a, b, d\}, \{a, c, d\}, \{a, b, c, d\}, \{b, d\}, \{b, c, d\}, \{c, d\}$.

Check Your Progress 4

- 1) a) $X \cup Y = \{1, \{2, 3\}, \{4, 5\}, 6, \{2, 4\}, \{5, 6, 7\}\}$;
b) $X \cap Y = \{1\}$;
c) $X - Y = \{\{2, 3\}, \{4, 5\}, 6\}$
d) $Y - X = \{\{2, 4\}, \{5, 6, 7\}\}$

Check Your Progress 5

- 1) $1.18725.4625 = 187254625 / 10000 =$ simplify by dividing numerator and denominator each by 25, and then dividing each of the resultant numerator and resultant denominator again by their G.C. D. (Greatest Common Divisor).
- 2) $2.31 / 12 = 2.58333... = 2.58 \bar{3}$

1.9 EXERCISES

Q1. Find the power set of the set $X = \{1, \{2, 3\}\}$

Ans. The set $X = \{1, \{2, 3\}\}$ has 2 elements. Therefore $P(X)$, the power set, has 4 elements. $P(X) = \{\Phi, \{1\}, \{\{2, 3\}\}, \{1, \{2, 3\}\}\}$.

Q2. State and prove De Morgan's Laws.

Ans. The two De Morgan's Laws about operations on sets, in English, state

- The complement of the union of two sets is the same as the intersection of their complements
- The complement of the intersection of two sets is the same as the union of their complements

In Mathematical Language

$$(A \cap B)^c = A^c \cup B^c \dots \dots (i)$$

$$(A \cup B)^c = A^c \cap B^c \dots \dots (ii)$$

Proof of (i) Set equality is proved by taking an arbitrary element of LHS, and showing that the element belongs to RHS, and vice-versa.

Let $x \in (A \cap B)^c$,

By definition of complement

$\Leftrightarrow x \in U$, the universal set, *and* $x \notin A \cap B$, (*The symbol $\Leftrightarrow$ stands for 'if and only if'*)

By definition of ' $\cap$ '

$\Leftrightarrow x \in U$ **and** ($x \notin A$ **or** $x \notin B$)

$\Leftrightarrow (x \in U \text{ and } x \notin A)$ **or** ($x \in U \text{ and } x \notin B$)

By definition of complement

$\Leftrightarrow (x \in A^c)$ *or* ($x \in B^c$)

By definition of union of sets

$\Leftrightarrow x \in A^c \cup B^c = \text{RHS.}$

Similarly,

Proof of (ii)

Let $x \in (A \cup B)^c$,

By definition of complement

$\Leftrightarrow x \in U$, the universal set, *and* $x \notin A \cup B$,

By definition of ' $\cup$ '

$\Leftrightarrow x \in U$ *and* ($x \notin A$ **and** $x \notin B$)

$\Leftrightarrow (x \in U \text{ and } x \notin A)$ **and** ($x \in U \text{ and } x \notin B$)

By definition of complement

$\Leftrightarrow (x \in A^c)$ *and* ($x \in B^c$)

By definition of intersection of sets

$\Leftrightarrow x \in A^c \cap B^c = \text{RHS.}$

UNIT 2 OVERVIEW OF BASIC METHODS OF MATHEMATICS

Structure

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2.0 OBJECTIVES

After going through this unit, you should be able to:

- explain the concepts of symbol, parameter, variable and constant;
- distinguish between, symbol, parameter, variable and constant;
- explain the concepts of mathematical expression/formula/well-formed-formula, equation, solution/root of equation, zero of function, polynomial identity, polynomials, and polynomial equations;
- use of polynomial identities in solving problems;

- apply algebraic operations to (i. e. add, subtract, multiply) polynomials; and
- solve linear equations in one variable, single linear equations in two variables, system of two or more linear equations in two or more variables, quadratic equations.

2.1 INTRODUCTION

In the previous unit, we discussed the two *basic concepts* of Mathematics, viz. *number* and *set*. In this unit, we discuss some *basic methods* for solving some types of equations. Mathematical *equations* involve concepts of *variable*, *constant*, *parameter* and *solution*. In this unit, before discussing the methods, we briefly discuss the concepts first.

In the *Appendix*, we discuss concepts and proof methods for deriving correct conclusions. A mathematical *proof* involves concepts of *hypothesis*, *conclusion*, *rules of inference* etc. You can learn how to apply various proof techniques to draw valid inferences from some given/ assumed facts.

2.2 CONCEPTS AND METHODS FOR SOLVING EQUATIONS

From our earlier encounters with solving equations, we know that $x^2 - 16 = 0$ is an *equation*, more precisely a *quadratic* equation, which involves one *variable* viz., x , and two *constants*, viz. 16 and 0. The equation has two *solutions*, viz., $x = 4$ and $x = -4$.

Before we discuss the topic in more details, it may be pointed out that equations play an important role in economics. Some specific equations, which actually state some law/ principle of economics, are quite well-known. For example, the *Fisher equation* expresses the relationship between nominal interest rates and real interest rates under inflation. it can be expressed as *real interest rate* $\approx$ *nominal interest rate* $-$ *inflation rate*, where the symbol $\approx$ denotes ‘approximately equal to’.

Though, we know intuitively the concepts of constant, variable, equation and solution, yet, it is desirable to examine these and other related concepts in some detail.

2.2.1 Symbol, Parameter, Variable and Constant

Symbol: More explicitly, each of the concepts viz. *constant*, *variable*, and *parameter* is a *symbol*, where, in general, a *symbol* is a notation which represents, or stands for, something else. For example, the four-letter word *lake* represents some water-body. By ‘something else’, we mean that through trying to touch the four-letter word *lake*, our finger does not get wet, but through trying to touch *what is represented* by the four-letter word, finger becomes wet. However, in case we wish to talk about the four-letter word *lake* itself, and not the water-body, then the *symbol* for it is generally written as either of (i) ‘lake’, (ii) “lake”, and (iii) *lake*, or some combination of these, e. g. ‘lake’.

Parameter: In addition to the equations of the type $x^2 - 16 = 0$, we also come across statements of the form: the equation $x^2 + y^2 = c$, with real number $c > 0$, represents a **family** of circles of radii $\sqrt{c}$ centered at origin $(0, 0)$. Here the c on R. H. S. is neither a constant nor a variable, but is a **parameter**, a concept somewhat a **hybrid** of constant and variable. By taking a *particular* constant value of c , say 4, we get *one* member, viz. the circle $x^2 + y^2 = 4$ of radius 2. Then, we may discuss various questions, e. g. dimensions of a rectangle, inscribed in the circle, having maximum area. Again, by taking $c = 15$, another constant, we pass on to *another* member of the family, viz. $x^2 + y^2 = 15$, a circle of radius $\sqrt{15}$. The **significance of the concept of parameter** lies in the fact that many questions like ‘dimensions of a rectangle, inscribed in the circle, having maximum area’ need **not** be discussed for each **individual** circle, but may be discussed for the **general** member/circle $x^2 + y^2 = c$, for arbitrary positive real number c . Then the answer for an individual case is obtained easily from the answer for the general case. In this sense, **a parameter is a sort of generalized constant**.

Variable is a symbol which *can be measured and which may change*. Examples of a variable from the field of economics include price, interest-rate, wages, quantity-bought, and quantity-sold. The quantity associated with a variable at some particular point of time is called its **value** at that particular point of time. For example, for the variable *Interest-rate* on a particular day, if it is 6%, then 6% is a **value** of ‘Interest-rate’. A variable, e. g. ‘Interest-rate’, may *assume only* any of its possible values. For example, $(\sqrt{-8})$ is not a possible value for the variable ‘percentage-of-Interest-rate’; but it may be a *possible value* for the variable x in the equation $ax^2 + bx + c = 0$. The end letters x, y, z of Latin alphabet along with some other letters are *commonly* used for naming variables.

Constant is a symbol for which *possible value* is a unique *single* value. For example, each of the symbols 6 in Latin notation, and ६, in Devnagari notation, can assume only a unique value. Generally, the beginning letters a, b, c etc. of Latin alphabet along with some other letters are used for naming constants.

2.2.2 Some Algebraic Concepts and Expressions

An **equation**—for example, $ax^3 + bx^2 + c = dx^2 + g$, is a mathematical **expression** which involves the relation of equality ($=$) exactly once, and each of the L.H.S. and R.H.S. of $=$ is also a mathematical expression which does not involve any of the mathematical relations including $=, \neq, <, \leq$ and $\geq$ etc. We will discuss the concept of *expression* in some detail, but before that we discuss some of the relevant issues, about which we know intuitively, yet require some formal statement.

2.2.2.1 Expression

Language is a means of communication between cohorts (for our purpose, who share some common minimum vocabulary and language-structuring rules). **Grammar of a language** is the set of all rules for structuring more complex linguistic entities from simpler ones, starting from the basic units, e. g. Latin

alphabet for English language. Thus, grammar informs us that the sequence ‘*Everest is the highest mountain peak in the world*’ of characters of Latin alphabet, is correct/ acceptable in English language, but the sequence ‘*the world in Everest peak highest mountain is the*’ is not acceptable. In a language, whatever is constructed according to its structuring rules—the totality of which is called its **syntax/grammar**—is an **expression** in the language.

2.2.2.2 Mathematical Expression

With mathematical expression we mean a *sequence of mathematical terms* (including constants, variables, parameters, relations and operations) formed by using the structuring rules of mathematics, for example,

- i) Each of constant, variable, parameter, *by itself* forms a mathematical expression. Thus, each of 7, x, c, 14.78 is a mathematical expression.
- ii) Each *unary operator* should be accompanied, at appropriate position, by a single operand which may be a constant, variable, parameter, or a mathematical expression not containing relational operators. Thus, each of the following is a mathematical expression: $\sqrt{8-y}$ (involving unary operator $\sqrt{}$ and single operand $(8-y)$); $(x+y)^2$ (involving unary operator $()^2$ and single operand $(x+y)$), $e^{x+2y-4z}$ (involving unary operator e and single operand $x+2y-4z$), $\log(3x+5)$ (involving unary operator $\log$ and single operand $3x+5$), $|y+56|$ (Involving a unary operator $| |$ and single operand $y+56$). **But none of the following is a mathematical expression:** (a) $8\sqrt{}$, (b) $x \log$, (c) x^e , and (d) $y| |$, (as operator is not in the right position in each of these); (e) $\sqrt{8 < y}$, (f) $(36x = 49)^2$ (as each involves a relational operator).
- iii) Each of the **binary operator** should be accompanied, at appropriate positions, **by two operands**, each of which may be a constant, variable, parameter, and/or a mathematical expression not containing relational operators. Thus, each of the following is mathematical expression: $(\sqrt{867} - e^{x+2y-4z})$ (involving binary operator minus $(-)$, and two operands $\sqrt{867}$ and $e^{x+2y-4z}$). Similarly, each of $(x+2y-4z)$, and $(\sqrt{8-y} \times (x+y)^2 + \log(x-e^z))$ is a mathematical expression. **But none of the following is a mathematical expression:** $\sqrt{8 < y} \times (x+y)^2$; $((36x = 49)^2 - 8\sqrt{})$ (each involves a relational operator, and the last also involves $8\sqrt{}$, which is not a mathematical expression)
- iv) Similarly, each *binary relation* should be accompanied, at appropriate positions, by two constants, variables, parameters, and/or mathematical expressions not containing relational operators. Thus, $(\sqrt{867} - e^{x+2y-4z}) < (\sqrt{8-y} \times (x+y)^2 + \log(x-e^z))$ is a mathematical expression, **but** $(\sqrt{867} - e^{x+2y-4z}) < (\sqrt{8-y} \times (x+y)^2) > \log(x-e^z)$ **is not a mathematical expression**, as the R.H.S of the binary operation ‘ $<$ ’ is a mathematical expression containing a relation, viz. ‘ $>$ ’. However, sometime we write $x < y < z$, as a shorthand for $x < y$ and $y < z$, and similarly each of $x \leq y < z$, and $x > y > z$ etc. is shorthand.

- v) Other Mathematical Expressions: (a) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ (b) $\frac{dy}{dx}$ (c) $\int (3x + 2) dx$. We will discuss in the units ahead.

2.2.2.3 Classification of Mathematical Expressions

For our purpose, we may classify the mathematical expressions as (i) arithmetic expressions (ii) polynomial expressions (iii) algebraic expressions (iv) general mathematical expressions. *In this sequence, each preceding expression is special case of all the succeeding ones.*

- i) An **arithmetic expression** is a special case of mathematical expression which *does not involve any variable*, e. g. $((\sqrt{-8 + 489}) \times (7! + 56)^2 + \log(155 - e^3))$, where e is the Euler's constant 2.718281828459045...
- ii) A **polynomial expression** is a mathematical expression which may involve variables only with nonnegative integer powers. For example, each of (a) $((\sqrt{-8 + 489}) \times (7! + 56)^2 + \log(155 - e^3))$, (b) $7x^3 - 5x + ((\sqrt{-8 + 489}) \times (7! + 56)^2 + \log(155 - e^3))$ and (c) $(x + y)^2$ is a polynomial expression, where
- does not involve any variable, or equivalently involves, say x^0 , with exponent of x being 0,
 - involves only one variable, and
 - involves two variables.

However, **none of**

- $\sqrt[3]{8-y}$ (which is equal to $(8-y)^{1/2}$)
- $\log x$
- e^y
- $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$
- $\frac{dy}{dx}$
- $\int (3x + 2) dx$

is a polynomial expression.

- iii) An **algebraic expression** is a mathematical expression which may involve variables only with powers of the form p/q , with p and $q \neq 0$, being integers. Thus, $\sqrt[3]{8-y}$ is algebraic expression, though not polynomial expression. Also $[\sqrt[3]{1+x^2} / \sqrt[3]{1-x^2}]$ is an algebraic expression.

As, we saw earlier, these are not polynomial expressions too.

2.2.2.4 General Forms

Coming back to **mathematical equations**, which are special forms of **mathematical expressions**, with each involving exactly one *equal-to* ($=$) relation, and no other relation. From each type of **expression**, we get that type of **equation** by putting equal to sign between two expressions of that type. For example, $(x + y)^2 = 3x + y$ is a **polynomial equation**, and $[\sqrt[3]{1+x^2} / \sqrt[3]{1-x^2}] = 8x - 9$ is an

algebraic equation, but not polynomial equation. *For our purpose*, in addition to **algebraic equations**, mathematical equations also include

- i) Equations involving logarithms, Euler's constant $e = 2.718281828459045\dots$, trigonometric functions etc.
- ii) Set-theoretic equations, e. g. $(X \cap Y)' = (X' \cup Y')$, where X and Y are sets. This is not merely an equation, but an **identity**,
- iii) Logical equations (*to be discussed later*)
- iv) Derivatives (Differential equations), integrals (Integral equations) (*to be discussed later*).

By our discussion above, general form of equation may be of the form

$$[\sqrt{(1+x^2)} / \sqrt{(1-x^2)}] + e^x - \log(x^2) = \sin(x)$$

which may be also re-written as

$$[\sqrt{(1+x^2)} / \sqrt{(1-x^2)}] + e^x - \log(x^2) - \sin(x) = 0,$$

in which the R.H.S is 0. As the above equation involves variable, viz. x , from our discussion later, the equation may be written, **in the general form**, as

$$f(x) = 0 \dots \dots (i)$$

where $f(x) = [\sqrt{(1+x^2)} / \sqrt{(1-x^2)}] + e^x - \log(x^2) - \sin(x)$, is a function of variable, viz. x .

Root/ Solution of an equation: for equation (i) above, a solution/ root is a constant value say, **a** of variable x , such that **$f(a)$, which is a constant, is 0.**

For example, for the equation

$$f(x) = x^2 - 4 = 0,$$

by taking $x = 2$, we get $f(2) = 2^2 - 4 = 4 - 4 = 0$. Thus, $x = 2$ is a root/ solution of the equation $f(x) = x^2 - 4 = 0$.

*In this case, we may also say that **2 is a ZERO of the function $f(x)$** (zero is spoken of the **function $f(x)$** , and NOT of the **equation $f(x) = 0$**)*

2.2.2.5 Mathematical Identity

Consider the following mathematical expressions, involving a single variable X

- i) $X^2 - 16 = 0$, and
- ii) $X^2 - 16 = (X + 4)(X - 4)$.

The expression (i) above, is *satisfied by only two values* of X , viz. $X = 4$, and $X = -4$. But the expression (ii) above *is satisfied by every* real, and even complex, number.

Take, for example, $X = 3$, then

$$\text{L.H.S. of (ii)} = (3)^2 - 16 = -7, \text{ and}$$

$$\text{R.H.S. of (ii)} = (3 + 4)(3 - 4) = -7.$$

The expression (i) above is an example of an **equation**, and the expression (ii) is an example of an **identity**.

Though, most of the time, the notation ‘=’ is used for both mathematical ‘**equation**’, and mathematical ‘**identity**’, yet, in specific cases, where the distinction needs to be emphasized, the notation ‘ $\equiv$ ’ is used instead. For example, the expression (ii) above, is expressed as

$$X^2 - 16 \equiv (X + 4)(X - 4).$$

2.2.2.6 Various Usages of ‘=’ in Mathematics

The following are some of the different usages of the symbol ‘=’

- In **equations**: find the value of x satisfying the equation $x^2 + 4x + 3 = 0$;
- In arithmetical **laws/ identities**: e. g. $(a + b)^2 = a^2 + b^2 + 2ab$, for all real numbers a and b ;
- if we need *to be more specific*, then we use ‘ $\equiv$ ’, e. g. $(a + b)^2 \equiv a^2 + b^2 + 2ab$;
- in **relations**: (i) $3 = 4 + 2$ return is *false*, $3 = 2 + 1$ return is *true*, for example, in

In conditional statements: ‘If $x = 7$, then ...’;

- In **assignments**: Let $x = 8$. To be more specific, in some programming languages like Pascal, the symbol ‘:=’ is used instead, e. g. $x := 8$;

- In **definitions**: For example,

i) Area = length $\times$ breadth;

ii) $(dy/dx)_{x=a} = \lim_{h \rightarrow 0} [f(a + h) - f(a)]/h$

To be more specific, the symbol $\stackrel{\text{def}}{=}$ is used, e. g.

i) Area $\stackrel{\text{def}}{=} \text{length} \times \text{breadth}$;

ii) $(dy/dx)_{x=a} = \lim_{h \rightarrow 0} [f(a + h) - f(a)]/h$.

Check Your Progress 1

- Which of the following is *not* a mathematical expression (M. E.)? Justify your answer.

(i) $\sqrt{|y + 56|}$ (ii) $\sqrt{(8 - y) \times (x + y)^2}$ (iii) $\log((8 < y) \times (x + y)^2)$ (iv) $\log(x + y)^2$ (v) $\log((x + y)^2)$.

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